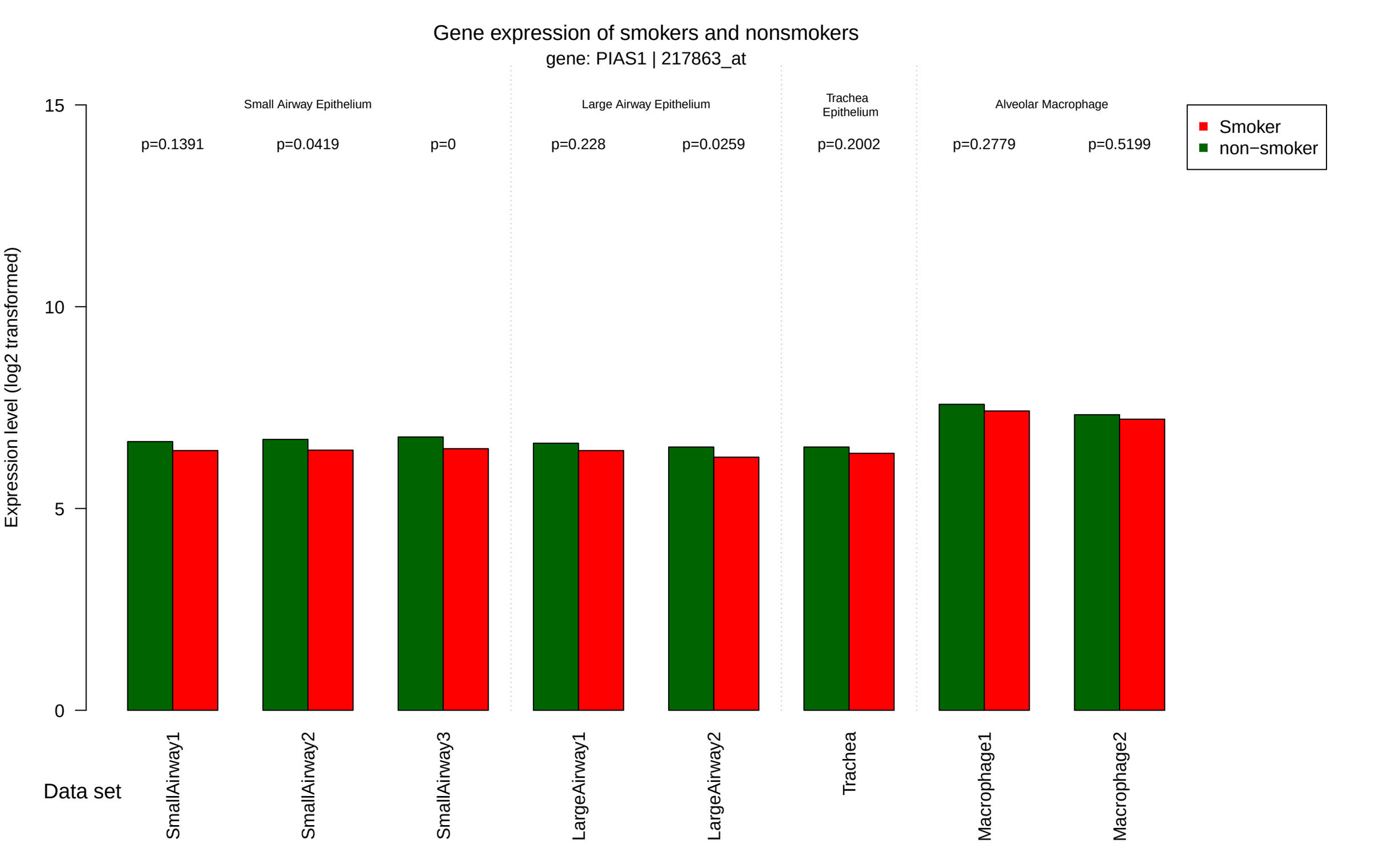
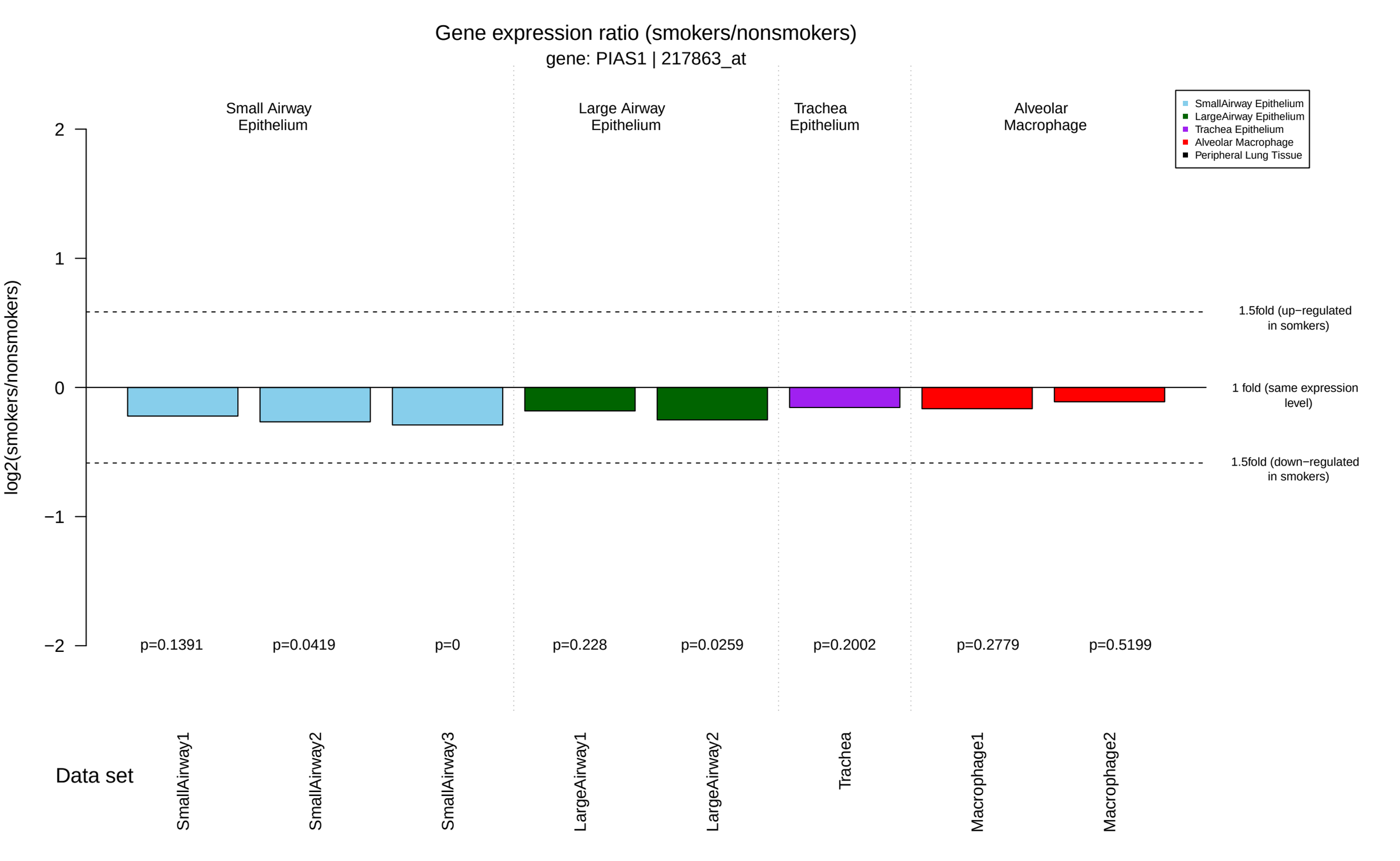
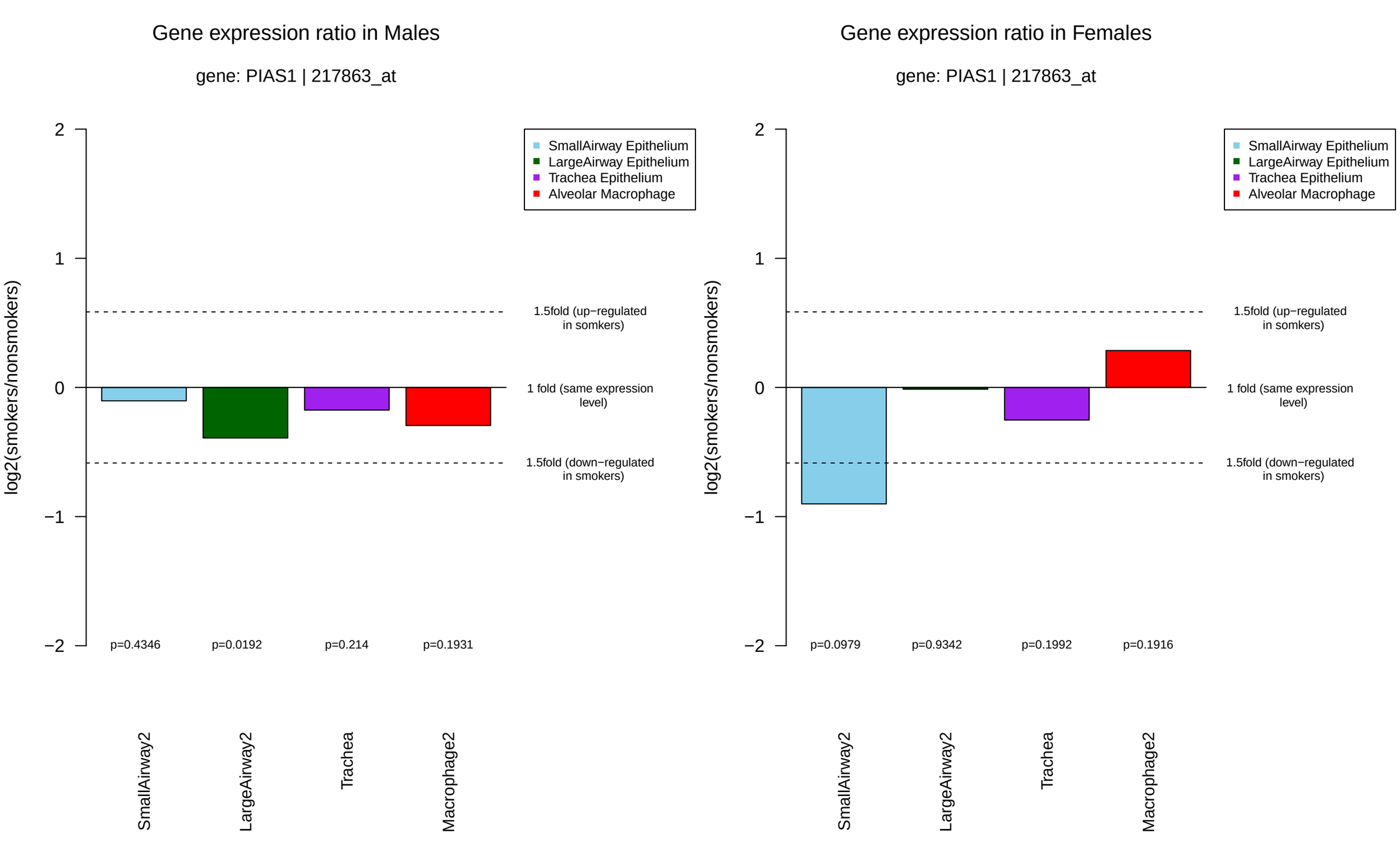
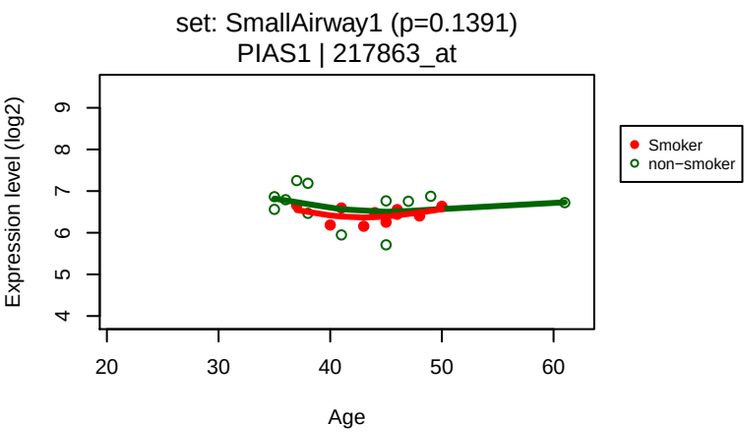
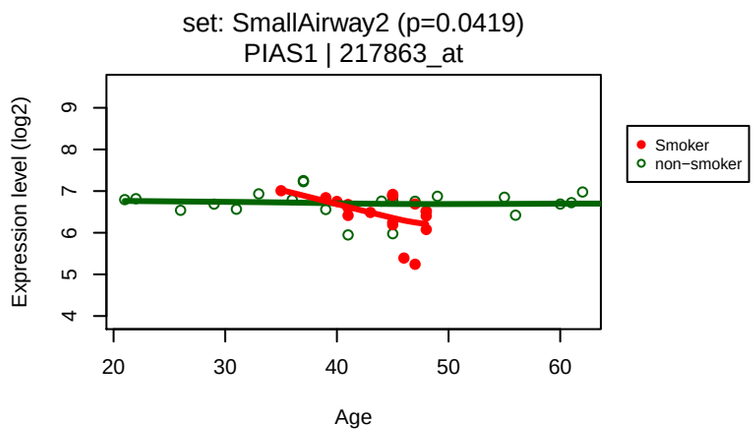
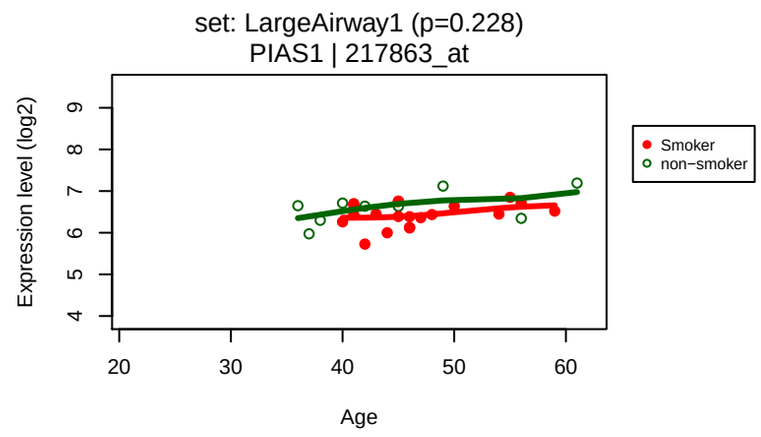

Gene expression of smokers and nonsmokers
gene: PIAS1 | 217863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1391
p=0.0419
p=0
p=0.228
p=0.0259
p=0.2002
p=0.2779
p=0.5199
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

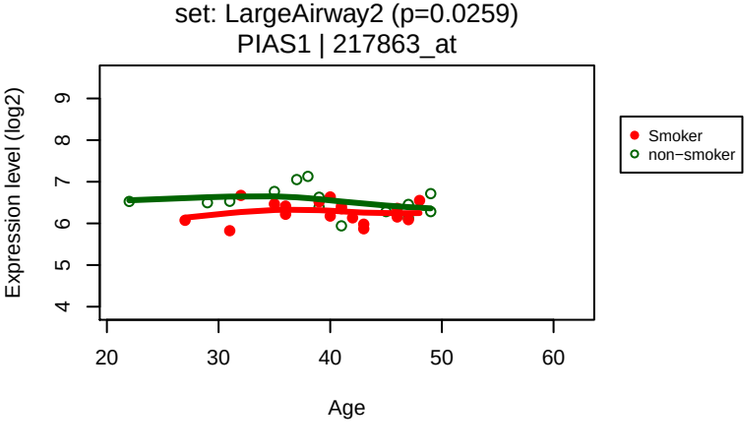

Gene expression ratio (smokers/nonsmokers)
gene: PIAS1 | 217863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1391
p=0.0419
p=0
p=0.228
p=0.0259
p=0.2002
p=0.2779
p=0.5199
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

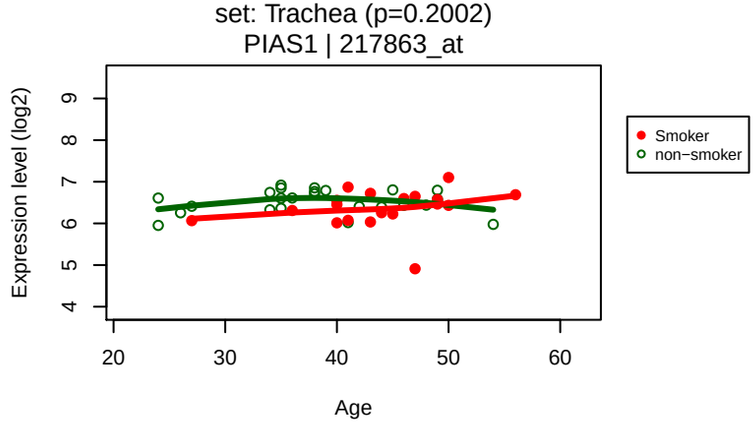

Gene expression ratio in Males
gene: PIAS1 | 217863_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4346
p=0.0192
p=0.214
p=0.1931
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIAS1 | 217863_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0979
p=0.9342
p=0.1992
p=0.1916
SmallAirway2
LargeAirway2
Trachea
Macrophage2

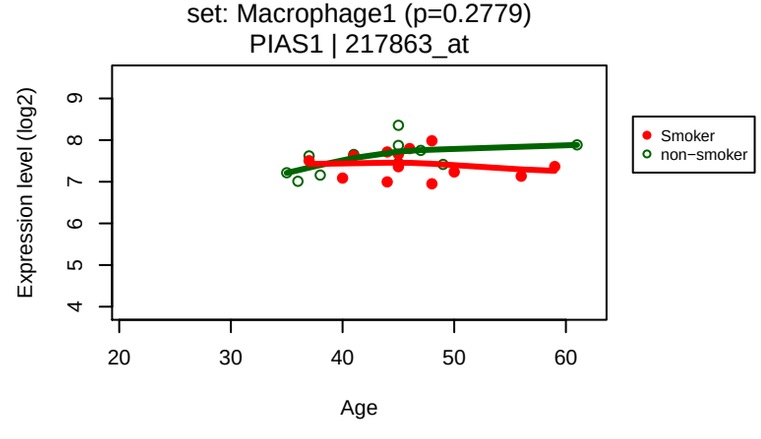

set: SmallAirway1 (p=0.1391)
PIAS1 | 217863_at
Expression level (log2)
Age
Smoker
non−smoker

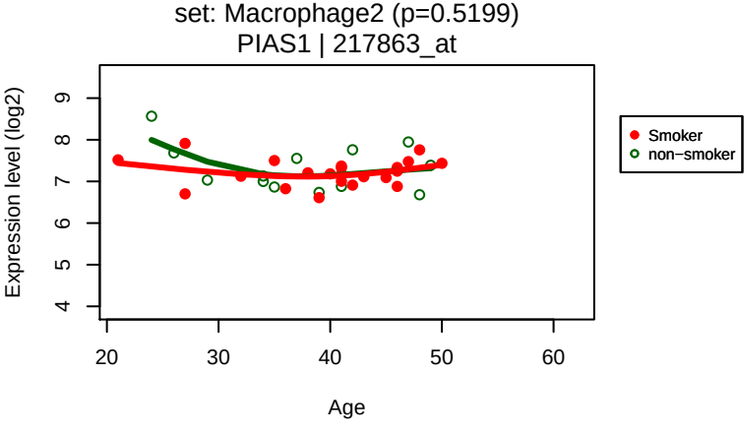

set: SmallAirway2 (p=0.0419)
PIAS1 | 217863_at
Expression level (log2)
Age
Smoker
non−smoker

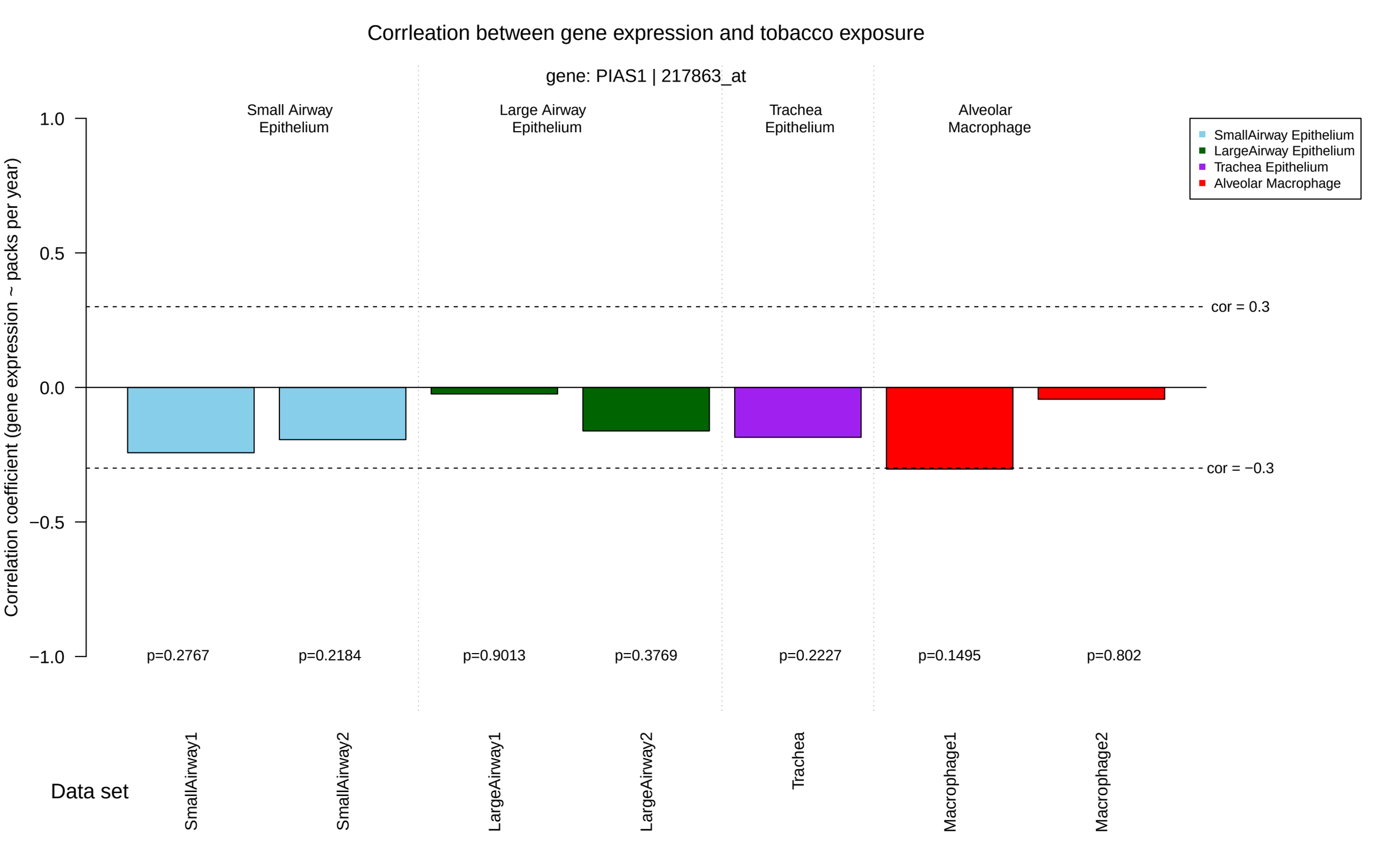

set: LargeAirway1 (p=0.228)
PIAS1 | 217863_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0259)
PIAS1 | 217863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2002)
PIAS1 | 217863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2779)
PIAS1 | 217863_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5199)
PIAS1 | 217863_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIAS1 | 217863_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2767
p=0.2184
p=0.9013
p=0.3769
p=0.2227
p=0.1495
p=0.802
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2